InfraSight

The AI co-pilot for city streets.

A citizen's photo becomes a mapped danger, an outlined damage, a picture of the repaired street, a merged case, a detour, a draft order and a message back home. Built on Meta's models.

- Muse Spark
- SAM 3.1
- Muse Image
- OpenStreetMap



THE PROBLEM

Every pothole is a small administrative maze.

A town hall can get hundreds of reports a year. Each one has to be understood, checked, routed, approved, bought, signed and closed, while citizens wait without news.



Unclear reports

Blurry photos, vague addresses, the same hole reported five times.



Who is responsible?

Commune, département, State, intercommunality, network operator or private owner.



Rules at every step

Purchase thresholds, council votes, DT-DICT, traffic orders, acceptance reports.



Silence for citizens

People rarely hear back, so they report again, or stop reporting.

BUILT ON META AI · USE OF META TOOLING

Three Meta models, one pipeline.

Muse Spark

Built · 9 AI functions

- → Reads photo and text: type, severity, owner, danger zone
- → Finds duplicates among nearby cases
- → Drafts the report, the document of each legal stage and the traffic order in French
- → Checks the repair: before vs after photo
- → Updates in 9 languages, weekly brief, forecast

→

SAM 3.1

Built · runs after every report

- → Outlines the exact shape of the damage on the citizen's photo
- → The outline becomes the red damage zone
- → It guides the size estimate and the repair preview

→

Muse Image

Built · repair preview

- → Draws the street after repair from the citizen's photo
- → Edits only the area SAM 3.1 marked in red
- → Shown as "expected" next to the real after-repair photo

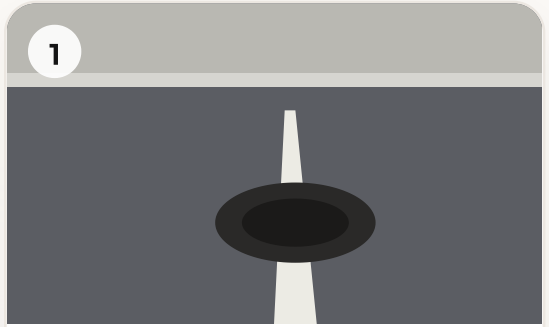
Meta Model API

All three run on Meta's servers with one API key, no GPU needed: about **$0.02 per citizen report**. Grounded in OpenStreetMap for addresses, street shapes and the road network.

SAM 3.1 · MUSE IMAGE · MUSE SPARK

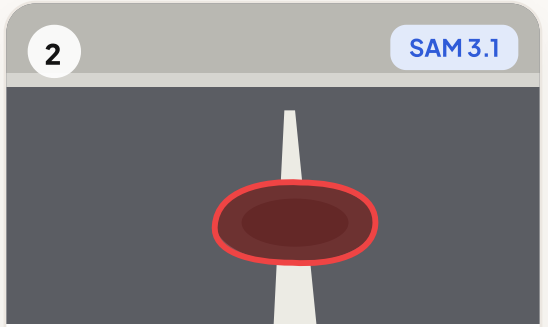
From a photo to the repaired street.

One photo, reused at every step. The same images follow the case everywhere: the ticket, the AI report, the map, the acceptance stage and the downloadable report.



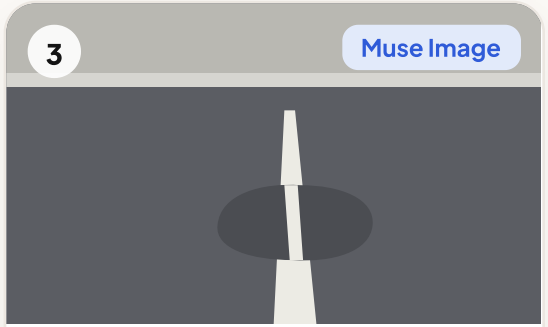
Citizen photo

Taken on a phone, pinned on the map. Muse Spark triages it.



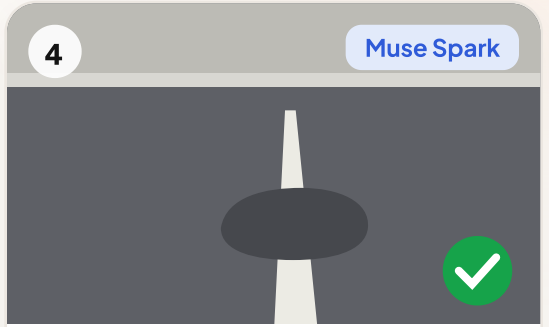
Damage zone

SAM 3.1 outlines the hole; Muse Spark measures it from the road marking and kerb.



Expected result

Muse Image repairs only the red area. Labelled as an illustration, not a photo.



Repair check

At acceptance, Muse Spark compares the real after photo with the red-marked "before".

Schematic illustration of the flow. Every AI output is tagged in the case history, and staff confirm on site.

TECHNICAL DEPTH · TRUSTWORTHY AI

Engineered to be checked, not just prompted.

Rules apart from data

Every call sends a system prompt (role, steps, rules) and the data in tagged blocks. Citizen text is declared untrusted, so it cannot steer the AI.

Written scales

Type, owner, severity 1–5, urgency 1–5 and danger-zone shapes are defined, so two similar reports get the same grade.

Cited sources

Every legal statement cites its article from the rules given to the AI, or is marked [verify]. Inventing a law or an amount is forbidden.

Validated answers

JSON answers are checked against schemas. An invalid answer is retried once, with the exact error given back to the model.

Humans decide

Every AI output is tagged "AI-generated" in the audit log. Staff correct it; only an elected official signs the decision stage.

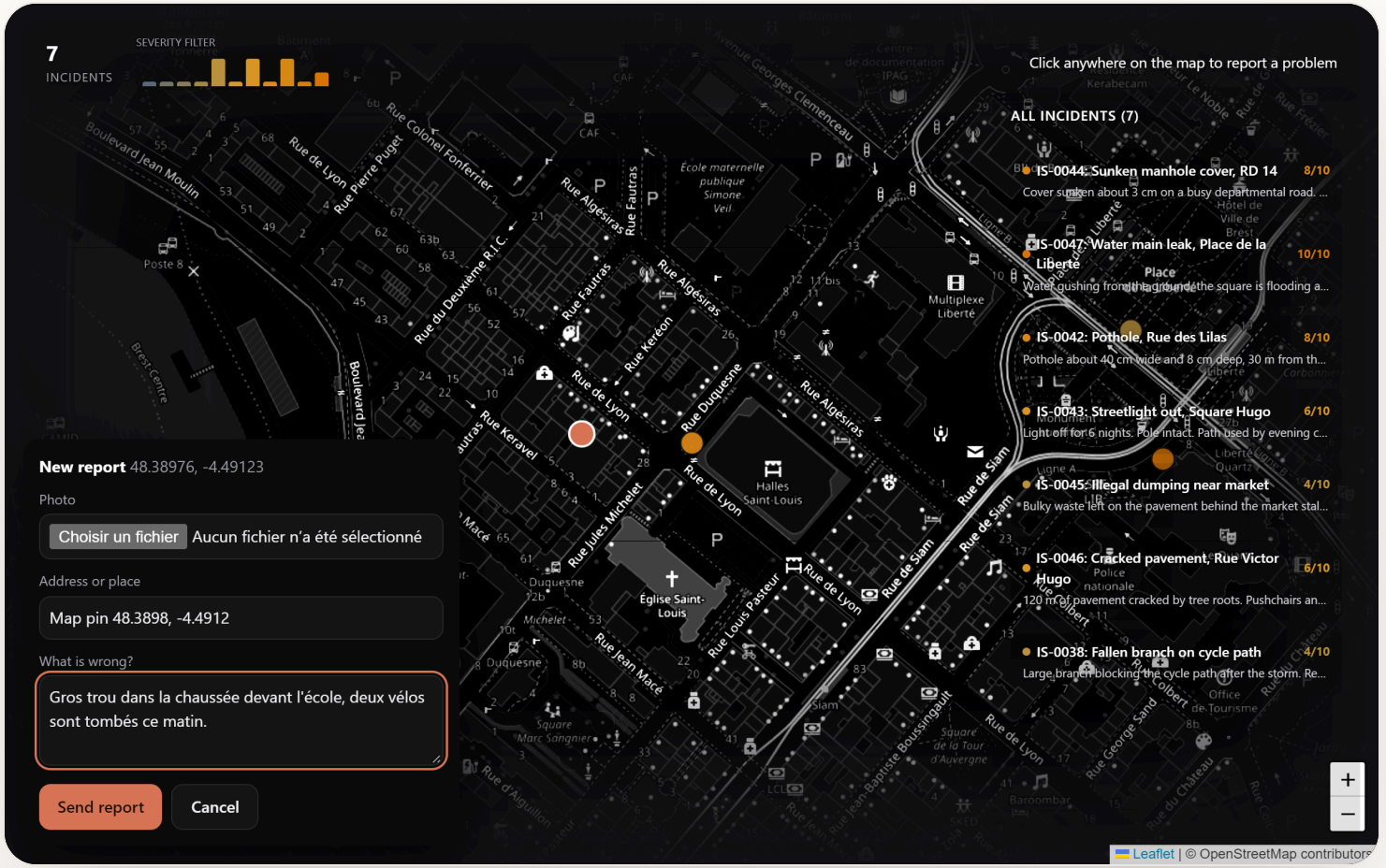
Robust

Prompts stay on the server, 30 AI calls per user per hour, accounts and roles. If the AI is down, the report is still saved.

FOR CITIZENS

Report in 30 seconds.

- Tap the spot on the map, the address fills itself.
- Add a photo and a few words, in your own language.
- Muse Spark classifies it and finds who is responsible.
- Updates arrive in the language you chose.

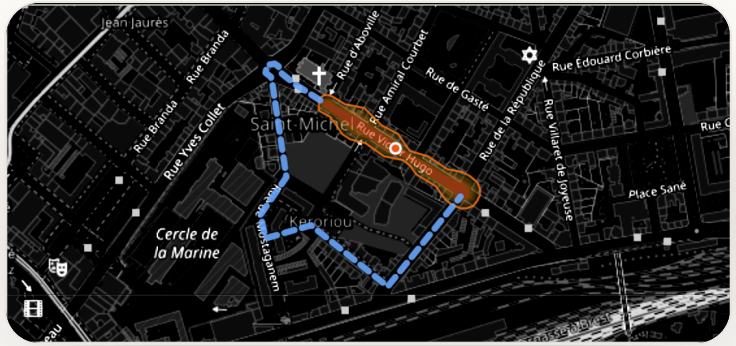


Screenshot of the working prototype, demo data in Brest

MUSE SPARK × OPENSTREETMAP

Danger zones shaped by the problem.

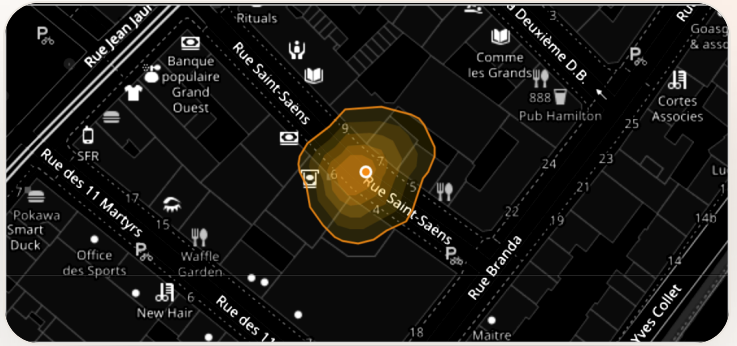
Muse Spark decides what kind of danger it is. InfraSight draws it on the real street: from yellow when the danger is low to red when it is severe or the road is blocked.



Blocked pavement · drawn along the whole street section



Water main leak · spreads over the square



Streetlight out · a small local zone

Danger — lower → higher

Zones are approximate: they guide action, they are not measurements. Each layer (reports, zones, detours, hotspots) can be shown or hidden on the map.

Screenshots of the working prototype

MUSE SPARK

Five reports, one case.

When a new report lands near an open case, Muse Spark is given the nearest cases with their distance and decides if it is the same problem. If so, the report joins the case: its photo goes in the history and the citizen follows the case.



Less work

One case to process instead of many.

Severity can rise

More reports and photos make a stronger case.

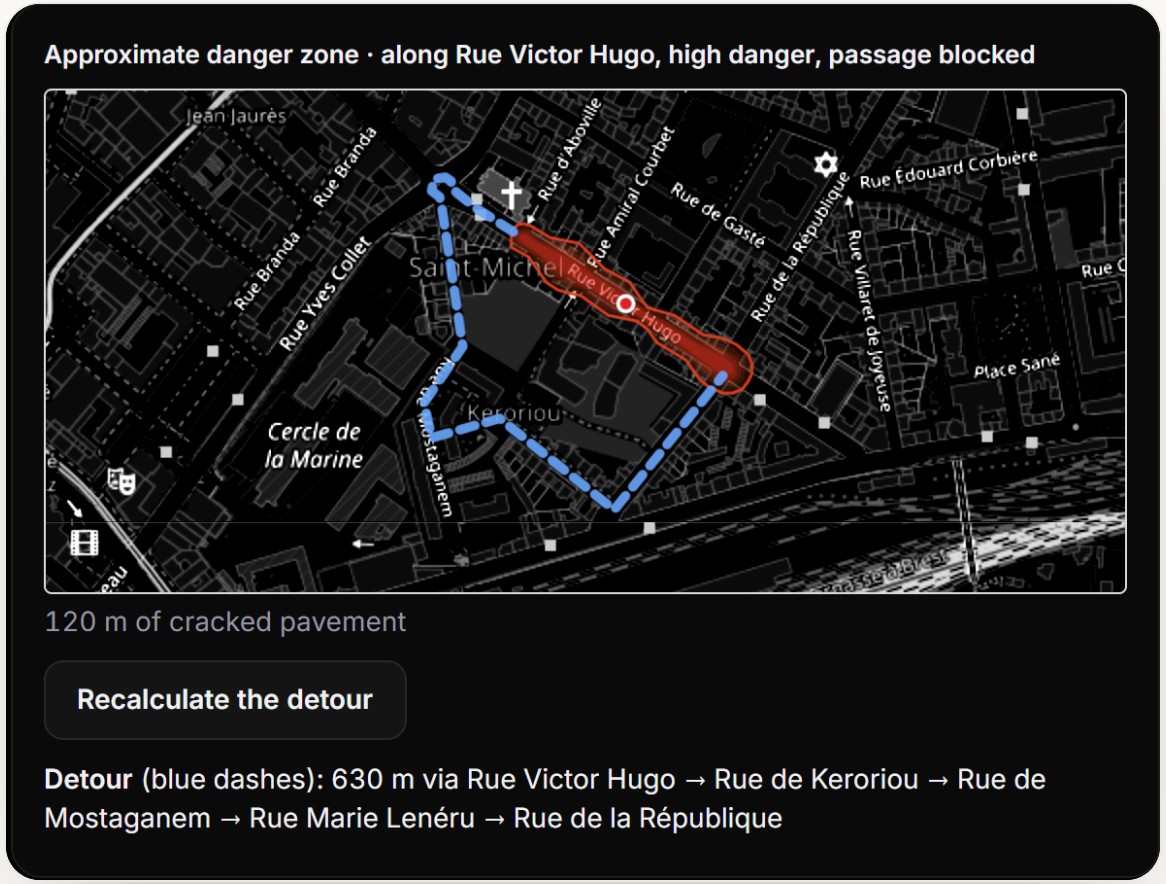
Everyone is told

All citizens who reported it get the same updates.

Screenshot of the working prototype

REAL DATA, REAL STREETS

The road is blocked. Here is the way around.



- InfraSight downloads the road network around the blocked section from OpenStreetMap.
- It computes the shortest legal detour, one-way streets included: here 630 m.
- Muse Spark drafts the mayor's traffic order in French, with the detour streets in it.
- Staff check it; the mayor signs.

Why it matters

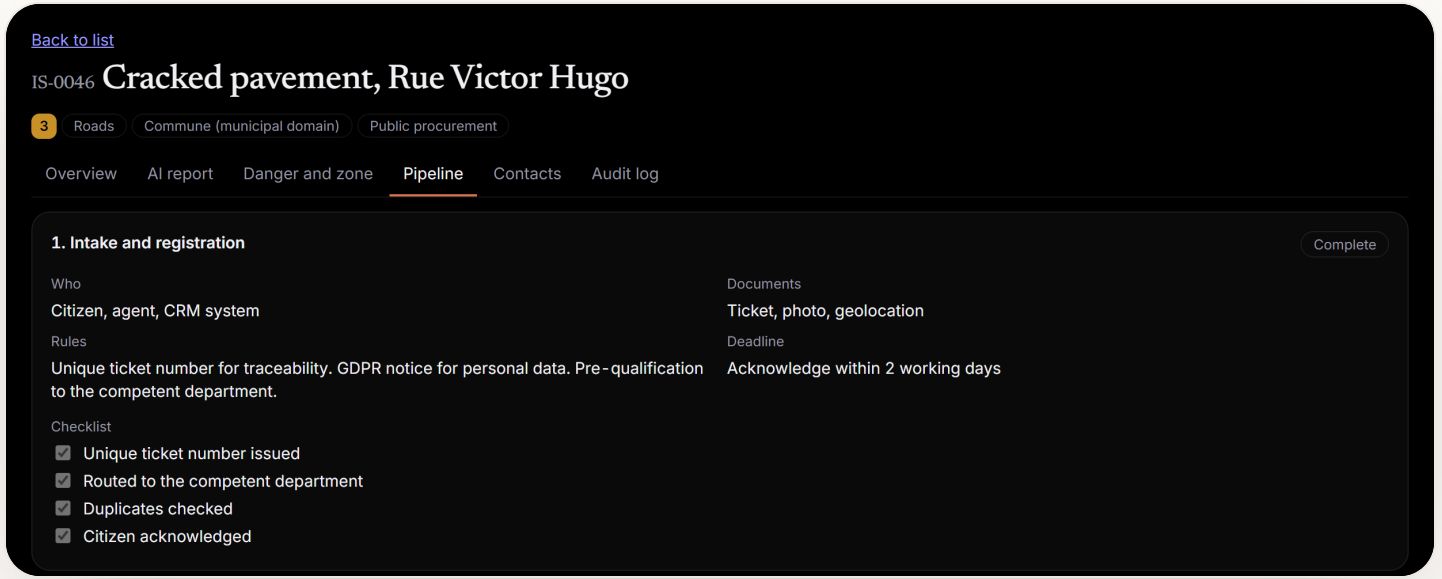
A traffic order normally takes hours of manual work: finding the detour, writing the articles, checking the legal basis.

Screenshot of the working prototype

MUSE SPARK FOR STAFF

The paperwork, drafted for you.

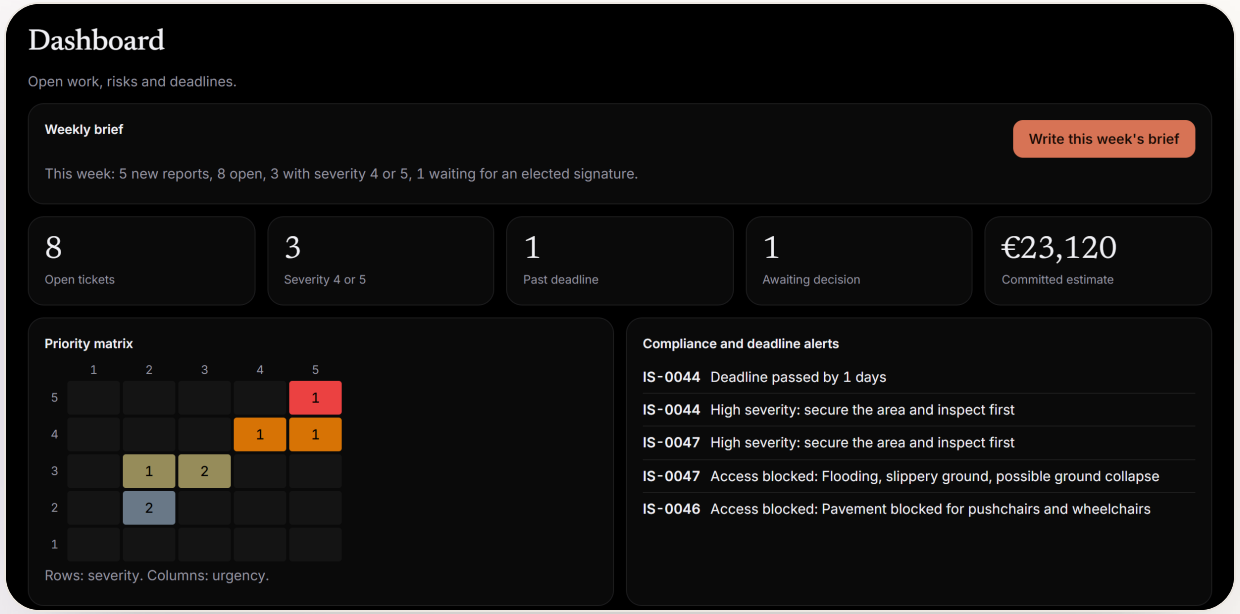
- 7 legal stages, from intake to acceptance, with the rules and deadlines of each.
- "AI assist" drafts the document of the current stage: decision memo, specifications, permits checklist, acceptance report.
- At closing, Muse Spark compares the before and after photos.
- Only an elected official can sign the decision stage.



Screenshot of the working prototype

FOR ELECTED OFFICIALS

See the whole city, act before it breaks.



Dashboard: priorities, alerts, and a weekly brief written by Muse Spark from the case data.

Repeated problems

Street	Reports	Open	Types	Avg sev.	Cost	Last
Avenue du Maréchal Foch	3	1	Roads 2, Water 1	3.7	€6,500	2026-11-01
Rue Victor Hugo	4	1	Roads 4	2.8	€25,850	2026-10-17

Hotspots: streets where problems keep coming back. Muse Spark turns them into a forecast with preventive actions and a budget.

CHECKED AGAINST THE HACKATHON'S ANTI-PROJECT LIST

Not a simple image analyzer. A public service, end to end.

The photo is where the work starts, not where it ends.

A SIMPLE IMAGE ANALYZER…	INFRASIGHT…
✗ labels a photo	✓ runs the case through 7 legal stages, from intake to acceptance and payment
✗ gives an answer in a window	✓ produces actions: danger zones on the map, merged cases, detours, draft orders, citizen messages
✗ calls one model once	✓ chains Muse Spark, SAM 3.1 and Muse Image, grounded in real road data from OpenStreetMap
✗ forgets the photo after one look	✓ reuses it at each step: triage, size, full report, repair preview, before / after check
✗ stops at a result	✓ keeps people in charge: three roles, an audit log, and an elected official's signature

Also not on the list: no medical advice, no politics, no basic RAG, no education or job chatbot. A tool for municipal services and the citizens they serve.

FOR THE JUDGES

Built for the four criteria.

Impact

30%

Every French commune handles road, lighting, waste and water reports, and citizens rarely hear back. InfraSight drafts the paperwork of each legal stage, routes each case to the right authority and keeps citizens informed in their language. Ready for a 3–month pilot.

Use of Meta AI tooling

25%

3 Meta models and 10 AI functions, chained: Muse Spark triages, SAM 3.1 outlines the damage, Muse Image previews the repair, Muse Spark checks it at acceptance. All on the Meta Model API, multimodal from the first step.

Technical depth

20%

FastAPI backend with roles and an audit log, schema-validated AI output, prompts with cited sources, OpenStreetMap routing for detours, and CI on every push.

Demo

25%

A working console for citizens, technicians and elected officials, on phone and desktop. The screenshots in this deck come from the working prototype, with demo data in Brest.

NEXT

From co-pilot to agent.

Next

InfraSight Agent

Muse Spark with tool calling handles a report end to end: looks at the photo with SAM 3.1, reads the road data, finds nearby cases, estimates the cost, checks the purchasing rules, plans the detour. Every step visible live; every change waits for a staff click.

Next

Learns from staff

Each correction by a technician is stored and reused. The dashboard shows how often the AI was right, and sends uncertain cases straight to a human.

Done during the hackathon

SAM 3.1 and Muse Image now run on the Meta Model API, with no GPU: every report gets an exact damage outline and a picture of the repaired street.

WHERE WE ARE

A working prototype, ready for a pilot.

- Citizen, technician and elected-official accounts
- Muse Spark triage, reports, stage documents, updates, forecast, brief
- SAM 3.1 damage outline and Muse Image repair preview
- Danger zones, duplicate merging, detours, hotspots

AFTER THE HACKATHON

A 3-month pilot with one commune

- Real reports from one neighbourhood
- Legal review of every rule built into the stages
- Measure response times and citizen satisfaction

Let's fix streets faster, together.